

Education

University of Illinois Urbana-Champaign

Bachelor of Science in Engineering Mechanics | Minor in Materials Science

May 2027

GPA: 3.86

Skills

- Analysis & Design: First Principles Analysis, GD&T, FEA (Ansys Mechanical/ACP), DFM/DFA, Structural Optimization.
 - Software: SolidWorks, PTC Creo, Ansys, CATIA V5, Abaqus, Python (Scripting for Automation), MATLAB & AutoCAD.
 - Fabrication & Testing: 3DP, Carbon Fiber Layup, CNC Machining, rapid prototyping, UTM (Tensile/Shear testing).
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Work Experience

Force Motors Ltd. | Automotive

Mechanical engineering Intern, R&D Powertrain

June 2025 to Aug 2025

- Developed a Python-based parametric design tool to automate critical crankshaft geometry sizing based on peak firing pressure loads, cutting design validation time by about 25%.
- Developed an analytical model to optimize counterweight distribution, achieving a >90% reduction in static and couple imbalance.
- Reverse-engineered 8 legacy 2D drawings into 3D models on CATIA V5 as part of a team-wide CAD data migration initiative, identifying tolerance stack-up issues and authored 6 production-ready engineering drawings.

Gala Precision Engineering | Coil Manufacturing

Mechanical engineering Intern

June 2024 to Aug 2024

- Conducted comparative FEA simulations (Ansys) on open vs. closed coil configurations, quantifying a 20% difference in compressive force to drive data-backed design tradeoffs.
 - Designed and deployed a mechanical conveyor modification (Kaizen) that eliminated part spillage during stress-relieving, reducing daily labor downtime by 15 minutes.
 - Validated spring performance through physical load testing, correlating displacement data with FEA models to within 12% accuracy.
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Project & Leadership Experience

Illini Electric Motorsport – FSAE | Aerodynamics

Front Wing Structures Project Lead

Aug 2024 to Present

- Spearheaded the cradle-to-grave lifecycle of the front wing assembly, leading 6 engineers through concept, design, FEA validation, manufacturing, and validation testing.
- Engineered a high-stiffness composite structure achieving <0.48" deflection at 95 mph validated using Ansys.
- Leveraged Topology Optimization on mounting struts to maximize strength-to-weight ratio, achieving an 18% increase in frontal impact energy absorption vs. legacy designs with zero mass penalty.
- Worked cross functionally with the VD and Chassis to define mounting interfaces and reach stiffness targets.
- Executed hands-on manufacturing of complex geometries, logging 100+ hours in pre-preg layup, vacuum bagging, and curing cycles to ensure manufacturing fidelity matched design intent.

UIUC Aerospace Engineering | Composites Research

Undergraduate Research Assistant

Jan 2025 to May 2025

- Quantified mechanical performance of a novel resin formulation for continuous carbon fiber composites, executing tensile/shear tests to determine material allowables (UTS: 1395 MPa).
 - Designed custom test coupons in Creo to overcome additive manufacturing constraints (minimum tow length), ensuring valid failure modes and reliable data capture.
 - Authored a technical paper detailing failure modes and resin performance for structural applications.
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Other Roles: Course Assistant (Thermodynamics, Statics, Mechanics) | ASME E-fest Lead | TEC Advisor & Lead Photographer